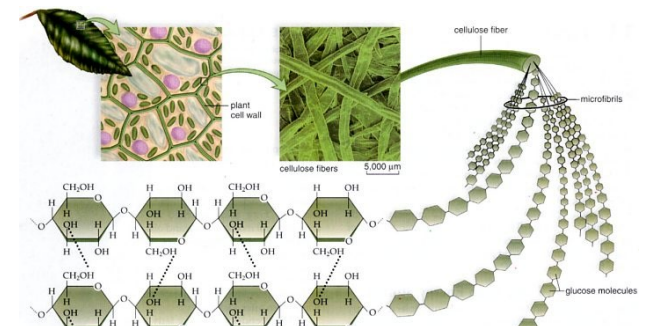


Nature Polymers

From page 84

TABLE 1 General Properties of Certain Natural Polymers

	Polymer		Incidence	Physiological function
A. Proteins	Silk		Synthesized by arthropods	Protective cocoon
	Keratin	角蛋白, 角朊	Hair	Thermal insulation
	Collagen	胶原蛋白	Connective tissues (tendon, skin, etc.)	Mechanical support
	Gelatin	明胶	Partly amorphous collagen	(Industrial product)
	Fibrinogen	纤维蛋白原	Blood	Blood clotting
	Elastin	弹力素	Neck ligament	Mechanical support
	Actin	肌动蛋白	Muscle	Contraction, motility
	Myosin	肌球蛋白	Muscle	Contraction, motility
B. Polysaccharides	Cellulose (cotton)	纤维素	Plants	Mechanical support
	Amylose	直链淀粉	Plants	Energy reservoir
	Dextran	葡聚糖	Synthesized by bacteria	Matrix for growth of organism
	Chitin	甲壳素	Insects, crustaceans	Provides shape and form
	Glycosaminoglycans	糖胺聚糖	Connective tissues	Contributes to mechanical support
C. Polynucleotides	Deoxyribonucleic acids (DNA)		Cell nucleus	Direct protein biosynthesis
	Ribonucleic acids (RNA)		Cell nucleus	Direct protein biosynthesis



Nature Polymers – general properties

Natural polymers offer the advantage of being very similar, often identical, to macromolecular substances which the biological environment is prepared to recognize and to deal with metabolically (Table 1). The problems of toxicity and stimulation of a chronic inflammatory reaction, which are frequently provoked by many synthetic polymers, may thereby be suppressed. Furthermore, the similarity to naturally occurring substances introduces the interesting capability of designing biomaterials which function biologically at the molecular, rather than the macroscopic, level. On the other hand, natural polymers are frequently quite immunogenic. Furthermore, because they are structurally much more complex than most synthetic polymers, their technological manipulation is much more elaborate. On balance, these opposing factors have conspired to lead to a substantial number of biomaterials applications in which naturally occurring polymers, or their chemically modified versions, have provided unprecedented solutions.

① 1

② 2

③ 3

④ 4

Non-toxic, great capability of designing new biomaterials

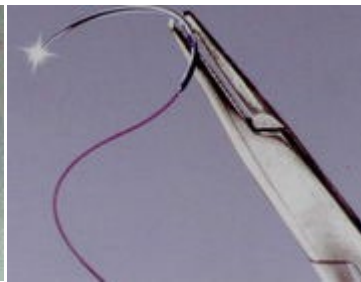
Nature Polymers – general properties

An intriguing characteristic of natural polymers is their ability to be degraded by naturally occurring enzymes, a virtual guarantee that the implant will be eventually metabolized by physiological mechanisms. This property may, at first glance, appear as a disadvantage since it detracts from the durability of the implant. However, it has been used to advantage in biomaterials applications in which it is desired to deliver a specific function for a temporary period of time, following which the implant is expected to degrade completely and to be disposed of by largely normal metabolic processes. Since, furthermore, it is possible to control the degradation rate of the implanted polymer by chemical cross-linking or other chemical modifications, the designer is offered the opportunity to control the lifetime of the implant.

① 1

② 2

③ 3



All organisms known to degrade cellulose efficiently produce a **battery of enzymes** with different specificities, which act together in synergism.

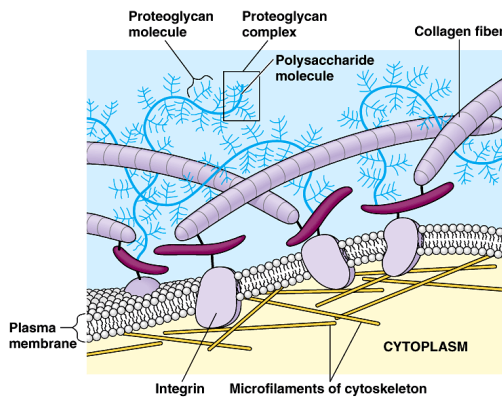
Nature Polymers - ECM

Most of the natural polymers in use as biomaterials today are constituents of the extracellular matrix (ECM) of connective tissues such as tendons, ligaments, skin, blood vessels, and bone. These tissues are deformable, fiber-reinforced composite materials of superior architectural sophistication whose main function in the adult animal appears to be the maintenance of organ shape as well as of the organism itself. In the relatively crude description of these tissues as if they were man-made composites, collagen and elastin fibers mechanically reinforce a “matrix” that primarily consists of protein-polysaccharides (proteoglycans) highly swollen in water. Extensive chemical

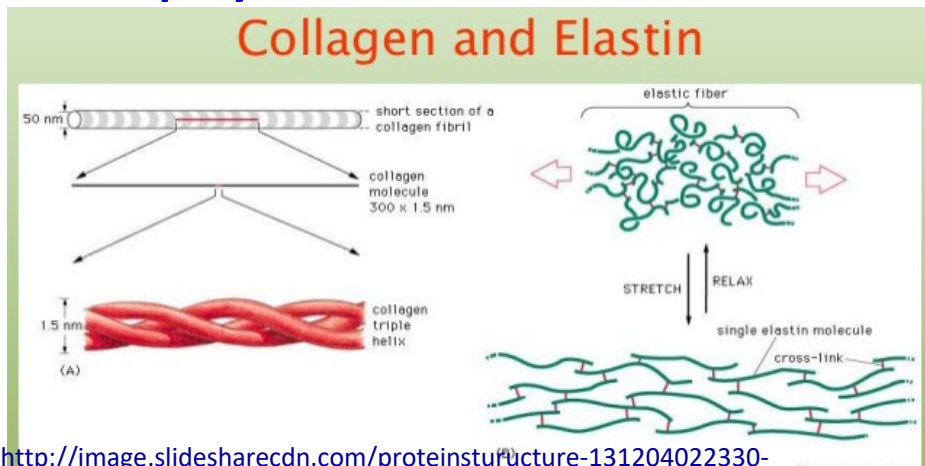
① 1

② 2

ECM: Collagen; Elastin; Protein – polysaccharides



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Liga
ment:
纽带;
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韧带

蛋白多
糖

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Nature polymers - native collagen

- ① 1 Structural order in collagen, as in other proteins, occurs at several discrete levels of the structural hierarchy. The collagen in the tissues of a vertebrate occurs in at least ten different forms, each of these being dominant in a specific tissue. Structurally, these collagens share the characteristic triple helix, and variations among them are restricted to the length of the nonhelical fraction, the length of the helix itself, and the number and nature of carbohydrate attachments on the triple helix.
- ② 2
- ③ 3 The collagen in skin, tendon, and bone is mostly type I collagen.
- ④ 4
- ⑤ 5 Type II collagen is predominant in cartilage, while type III collagen is a major constituent of the blood vessel wall as well as being a minor contaminant of type I collagen in skin. In contrast to these collagens, all of which form fibrils with the distinct collagen periodicity, type IV collagen, a constituent of the basement membrane which separates epithelial tissues from mesodermal tissues is largely nonhelical and does not form fibrils. We follow here the nomenclature which was proposed
- ⑥ 6

软骨

Where to get collagen? How many main types of collagen? Structure helical?

Structures of native collagen

It is believed that **collagen types I, II, and III compose the majority** of the body's collagen content. Of these three proteins, **collagen type I is believed to be the most abundant.**

Collagen I one can be found in many bodily structures, including bones, skin, and intervertebral disks. It is, therefore, present throughout the body. This type of collagen is commonly used in the gelatin industry and for research.

Collagen type III, which can be found in the skin, muscles, and intestinal walls, has been found to be the second most abundant type in the body. It is often closely associated with collagen I. This is because these collagen types are commonly found in the same places and they are synthesized in a similar manner.

Structures of native collagen

The primary structure denotes the complete sequence of amino acids along each of three polypeptide chains as well as the location of interchain cross-links in relation to this sequence. Approximately one-third of the residues are glycine and another quarter or so are proline or hydroxyproline. The structure of the bifunctional interchain cross-link is the relatively complex condensation product of a reaction involving lysine and hydroxylysine residues; this reaction continues as the organism matures, thereby rendering the collagens of older animals more difficult to extract.

① 1

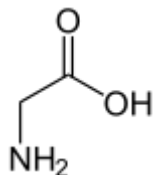
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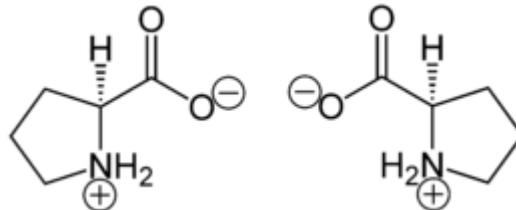
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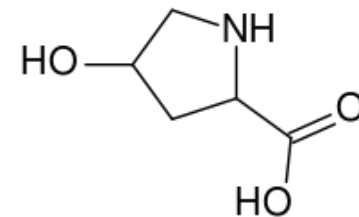
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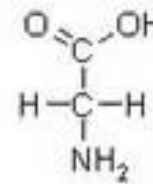
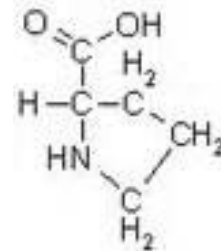
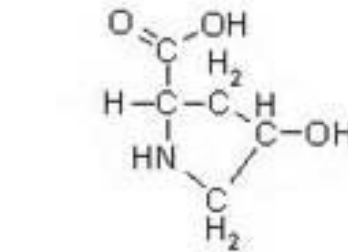
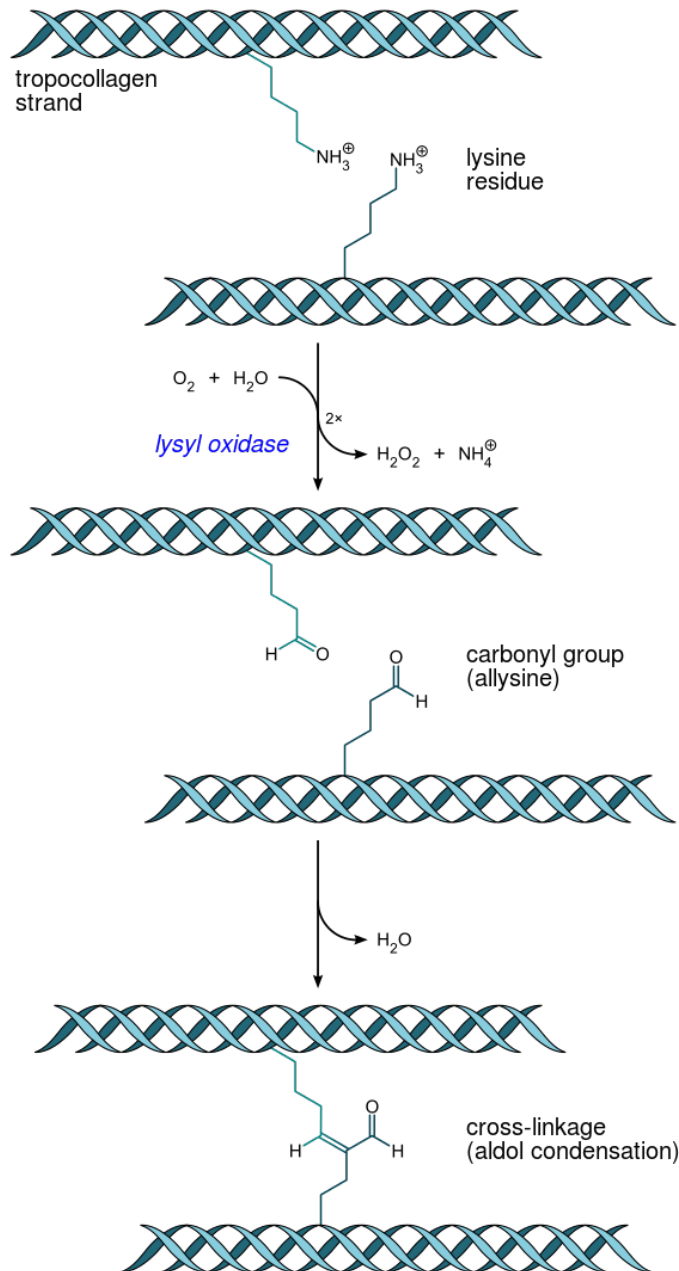
P 脯氨酸



Hyp

What are the key components of collagen? Why difficult to get collagen from older animals?

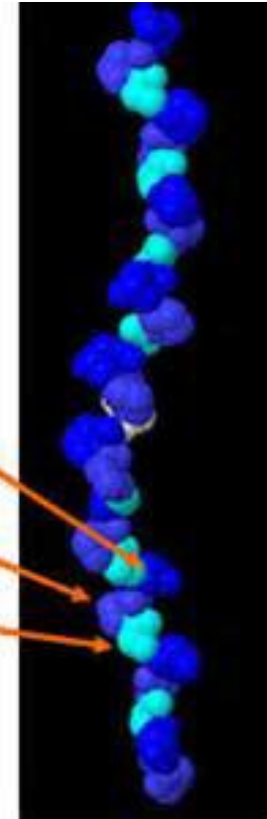
Structures of native collagen



Hydroxyproline

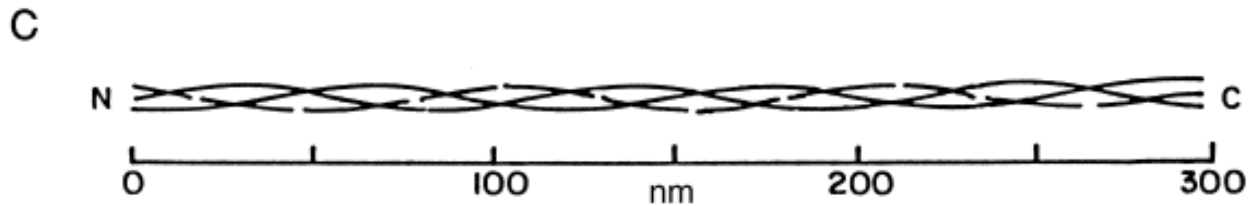
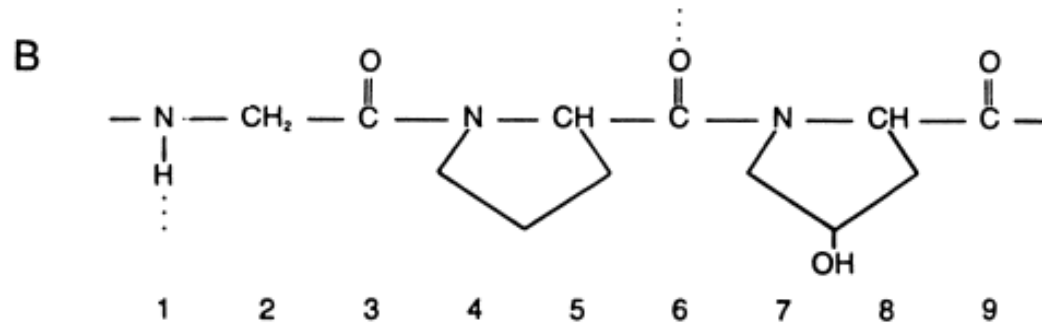
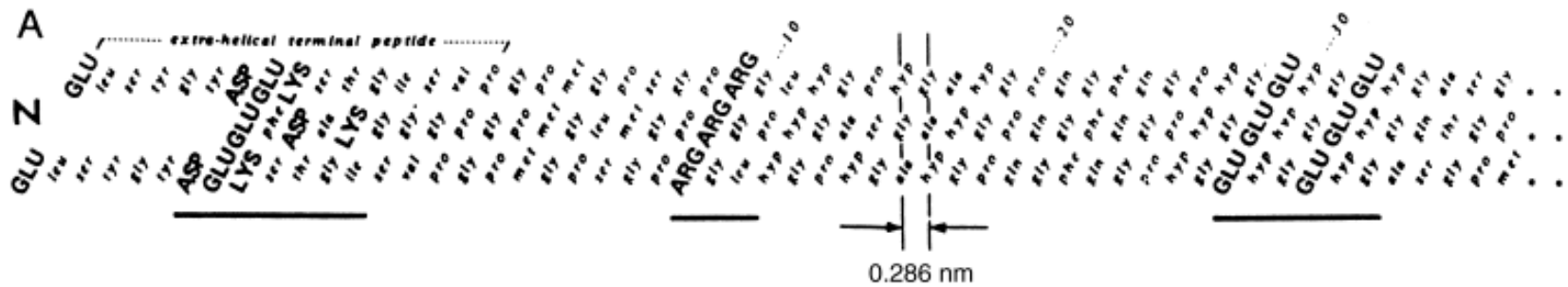
Proline

Glycine



What are the key components of collagen?
Why difficult to get collagen from older animals? Triple helix?

Structures of native collagen



Structures of native collagen

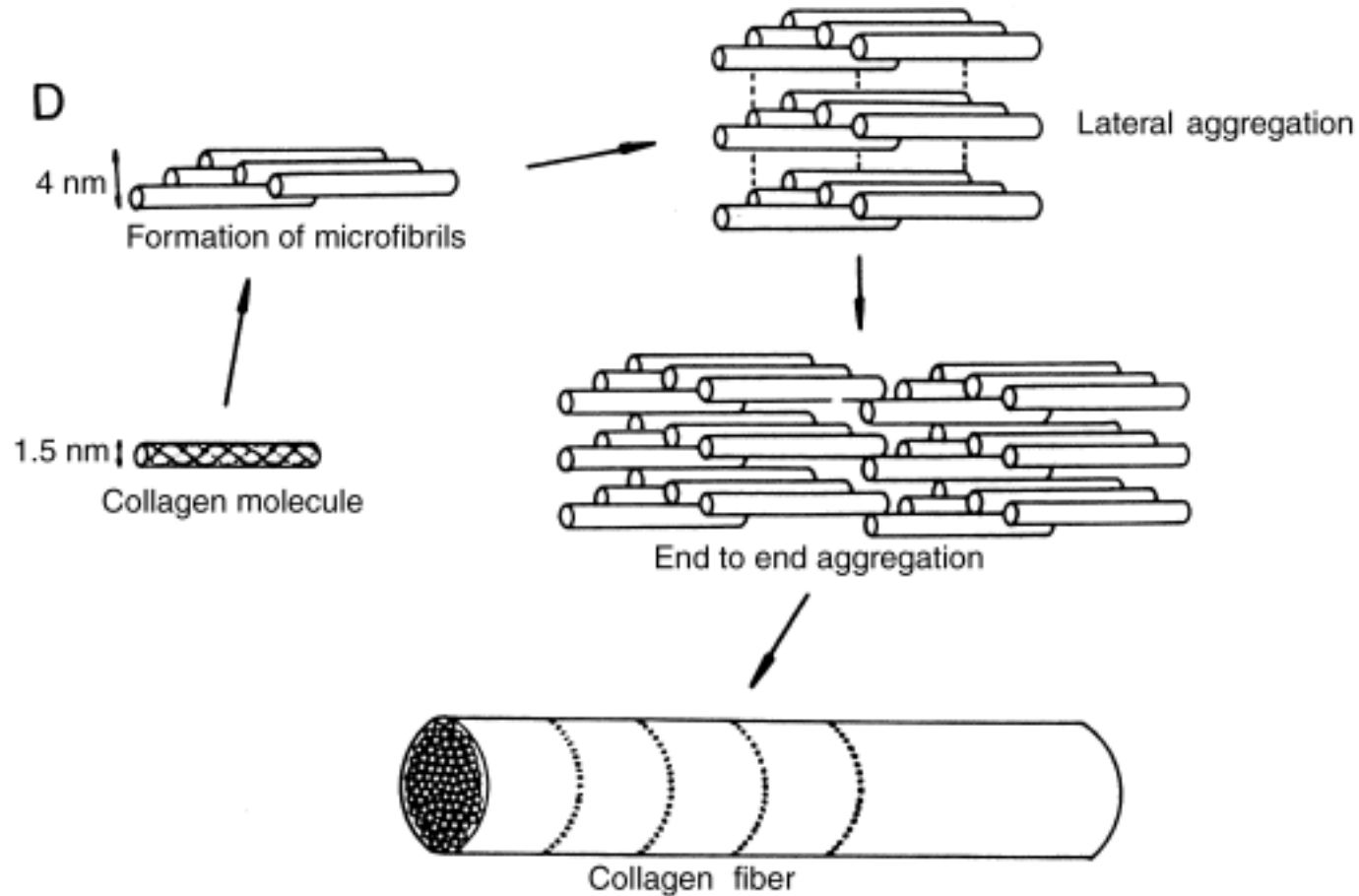


FIG. 1. For legend see opposite page.

Pitch of about 10 nm (30–40 residues).

Length of the helical portion of the molecule is **290 nm long**.

Packing in a specific lattice (**1.5 nm each**), which constitutes the basic element of the solid state (microfibril)

4 nm for one periodicity with a quasi-hexagonal lattice at an interchain distance of about 1.3 nm

Structures of native collagen

- ① 1 The secondary structure is the local configuration of a polypeptide chain that results from satisfaction of stereochemical angles and hydrogen-bonding potential of peptide residues. In collagen, the abundance of glycine residues (Gly) plays a key
- ② 2 configurational role in the triplet Gly-X-Y, where X and Y are frequently proline or hydroxyproline, respectively, the two amino acids that direct the chain configuration locally by the
- ③ 3 rigidity of their ring structures. On the other hand, the absence of a side chain in glycine permits the close approach of polypeptide chains in the collagen triple helix. The tertiary structure
- ④ 4 refers to the global configuration of the polypeptide chains; it represents the pattern according to which the secondary
- ⑤ 5 structures are packed together within the complete macromolecule and it constitutes the structural unit that can exist as a physicochemically stable entity in solution, namely, the triple
- ⑥ 6 helical collagen molecule.

Triple helical??

Structures of native collagen

In type I collagen, two of the three polypeptide chains have identical amino acid composition, consisting of **1056 residues** and are termed $\alpha 1(I)$ chains, while the third has a different composition, it consists of 1038 residues and is termed $\alpha 2(I)$. The three polypeptide **chains fold to produce a left-handed helix**, whereas the three-chain supercoil is actually right-handed with an estimated **pitch of about 10 nm (30–40 residues)**.

The helical structure extends over 1014 of the residues in each of the three chains, leaving the remaining residues at the ends (telopeptides) in a nonhelical configuration.

The residue spacing is **0.286 nm** and the length of the helical portion of the molecule is, therefore, 1014×0.286 or **290 nm long**.

Structures of native collagen

The fourth-order or quaternary structure denotes the repeating supermolecular unit structure, comprising several molecules packed in a specific lattice, which constitutes the basic element **of the solid state (microfibril)**.

Collagen molecules are packed in a quasi-hexagonal lattice at an interchain distance of **about 1.3 nm**, which shrinks considerably when the microfibril is dehydrated.

Adjacent molecules in the microfibril are approximately parallel to the fibril axis; they all point in the same direction along the fibril and are staggered regularly, giving rise to the well-known **D-period of collagen, about 64 nm**, which is visible in the electron microscope.

Higher levels of order, eventually leading to gross anatomical features that can be readily seen with the naked eye, have been proposed, but there is no general agreement on their definition.

D-period of collagen

Chains



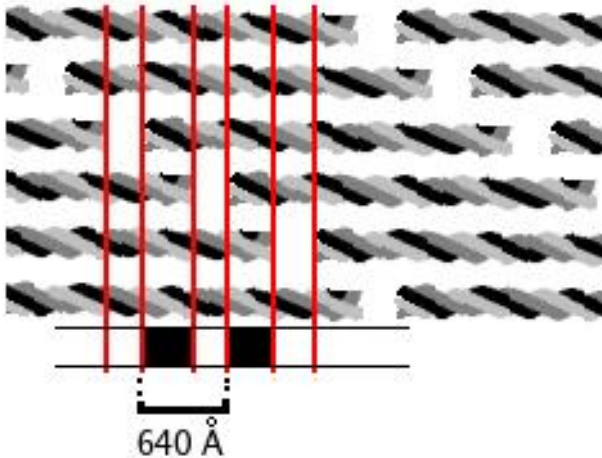
Procollagen Procollagen: 原骨胶原



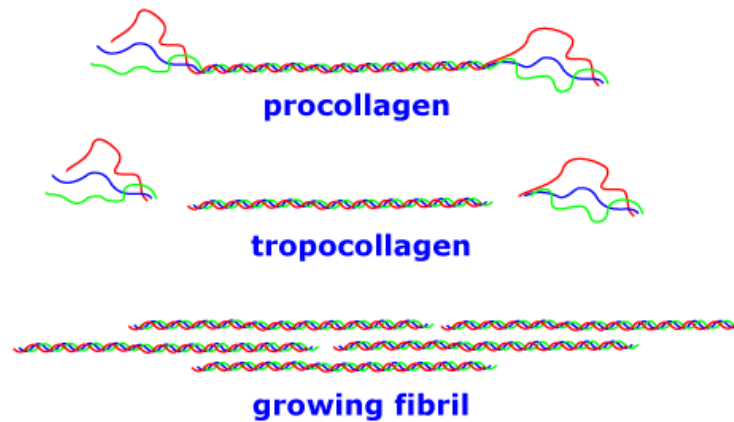
Tropocollagen 溶胶原



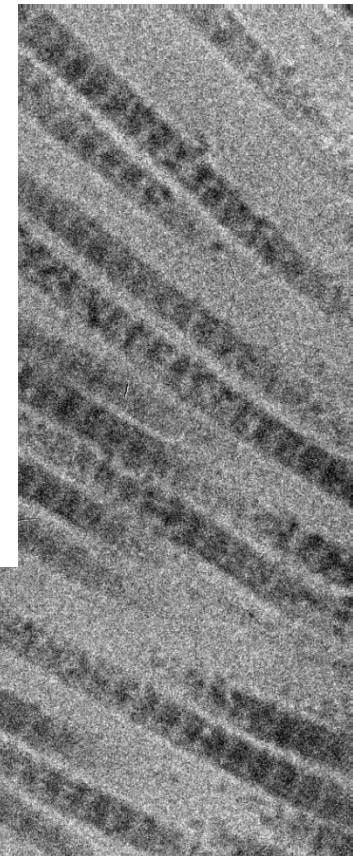
Fibril



Fibre



representation is schematic rather than actual



The D-period of collagen fibrils results in visible 67nm bands when observed by electron microscopy.

BIOLOGICAL EFFECTS OF PHYSICAL MODIFICATIONS OF THE NATIVE STRUCTURE OF COLLAGEN

Crystallinity in collagen can be detected at two discrete levels of structural order: the tertiary (triple helix) (Fig. 1C) and the quaternary (lattice of triple helices) (Fig. 1D). Each of these levels of order corresponds, interestingly enough, to a separate **melting transformation**.

A solution of collagen triple helices is thus converted to the randomly coiled gelatin by heating above the helix–coil transition temperature, **which is approximately 37°C** for bovine collagen, or by **exceeding a critical concentration of certain highly polarizable anions, e.g., bromide or thiocyanate, in the solution of collagen molecules**.

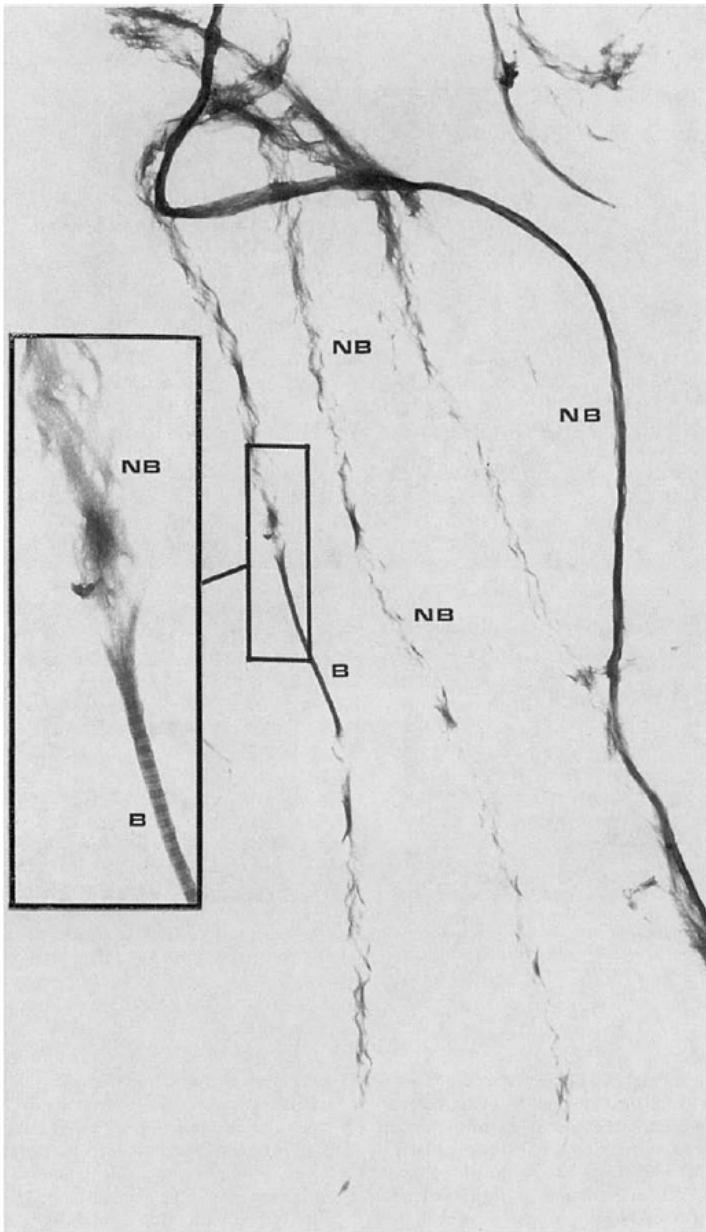
Collagen fibers also exhibit a characteristic banding pattern **with a period of about 65 nm (quaternary structure)**. This pattern is lost reversibly when the pH of a suspension of collagen fibers in acetic acid **is lowered below 4.25 ± 0.30** .

Structures of native collagen

Transmission electron microscopy or small-angle X-ray diffraction can be used to determine the fraction of fibrils that possess banding **as the pH of the system is altered**.

During this transformation, which appears to be a **first-order thermodynamic transition**, the **triple helical structure remains unchanged**. Changes in pH can, therefore, be used to **selectively abolish the quaternary structure while maintaining the tertiary structure intact**.

Following exposure to pH below 4.25 ± 0.30 the banding pattern of type I bovine hide collagen practically disappears. **Short lengths of banded collagen (B) do, however, persist next to very long lengths of nonbanded collagen (NB), which has tertiary but not quaternary structure.**



Structures of native collagen

The porosity of a collagenous implant normally makes an indispensable contribution to its performance.

A porous structure provides an implant with two critical functions.

First, pore channels are ports of entry for cells **migrating** from adjacent tissues into the bulk of the implant for tissue serum (exudate) that enters via capillary suction or of blood from a hemorrhaging blood vessel nearby.

Second, pores endow a material **with a frequently enormous specific surface** that is made available either **for specific interactions** with invading cells (e.g., myofibroblasts bind extensively on the surface of porous collagen–glycosaminoglycan copolymer structures that induce regeneration of skin in burned patients) or for interaction with coagulation factors in blood flowing into the device (e.g., hemostatic sponges).

Porous structures from collagen copolymers

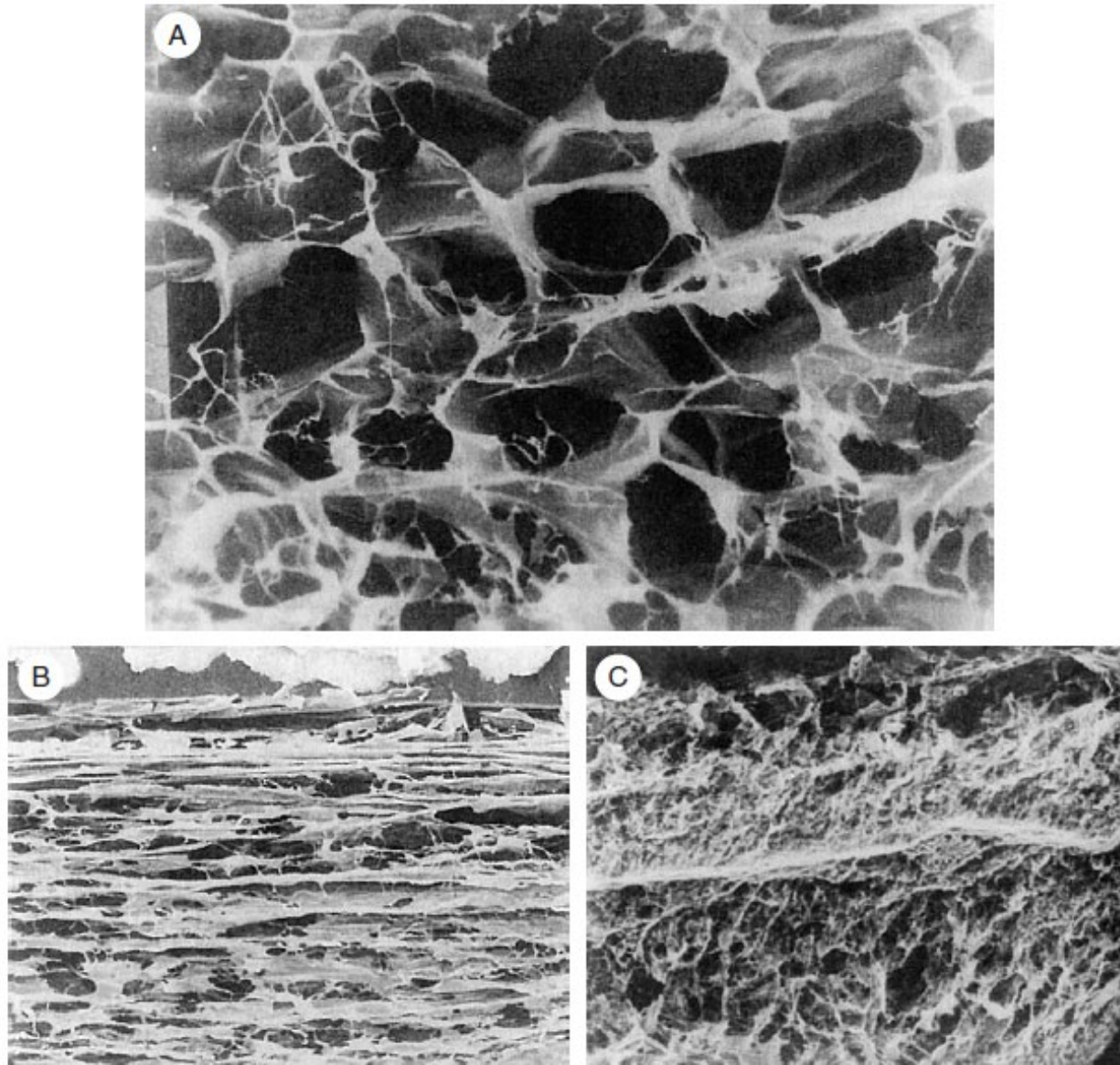


FIG. 3. Illustration of the variety of porous structures that can be obtained with collagen-GAG copolymers by adjusting the kinetics of crystallization of ice to the appropriate magnitude and direction. Pores form when the ice dendrites are eventually sublimated. Scanning electron microscopy. (Courtesy of MIT.)

Collagen – chemical modification

- Historically, the chemical modification of collagen has been
- ① 1 practiced in the leather industry (since about 50% of the protein content of cowhide is collagen) and in the photographic gelatin industry. Today, the increasing use of collagen in biomaterials applications has provided renewed incentive for novel chemical
 - ② 2 modification, primarily in two areas. First, implanted collagen
 - ③ 3 is subject to degradative attack by collagenases, and chemical
 - ④ 4 cross-linking is a well-known means of decelerating the degra-
 - ⑤ 5 dation rate. Second, collagen extracted from an animal source
 - ⑥ 6 elicits production of antibodies (immunogenicity). Although it
 - ⑦ 7 is widely accepted that collagen elicits synthesis of a far smaller
 - ⑧ 8 concentration of antibodies than other proteins (e.g., albumin),
 - ⑨ 9 treatment with specific reagents, including enzymatic treat-
 - ⑩ 10 ment, is occasionally used to reduce the immunogenicity of col-
 - ⑪ 11 lagen.

Why chemical modification?

Collagen – how degradation occurs?

Collagen-based implants are normally degraded by collagenases, naturally occurring enzymes which attack the triple helical molecule at a specific location. Two characteristic products result, namely, the N-terminal fragment which amounts to about two thirds of the molecule, and the one-quarter C-terminal fragment. Both of these fragments become spontaneously transformed (denatured) to gelatin at physiological temperatures via the helix-coil transition and the gelatinized fragments are then cleaved to oligopeptides by naturally occurring enzymes which degrade several other tissue proteins (non-specific proteases).

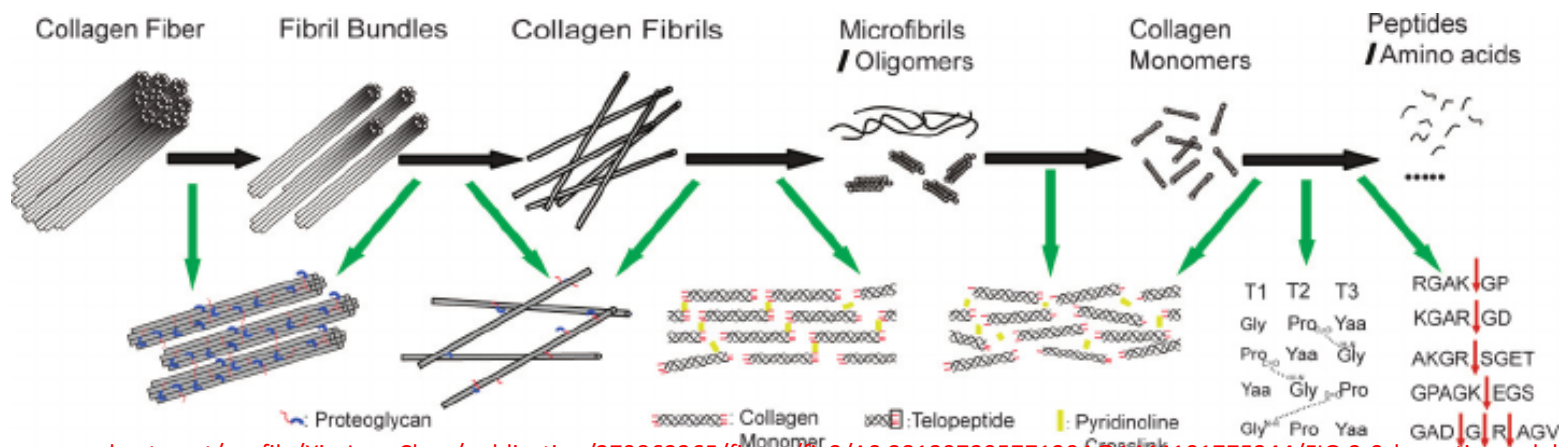
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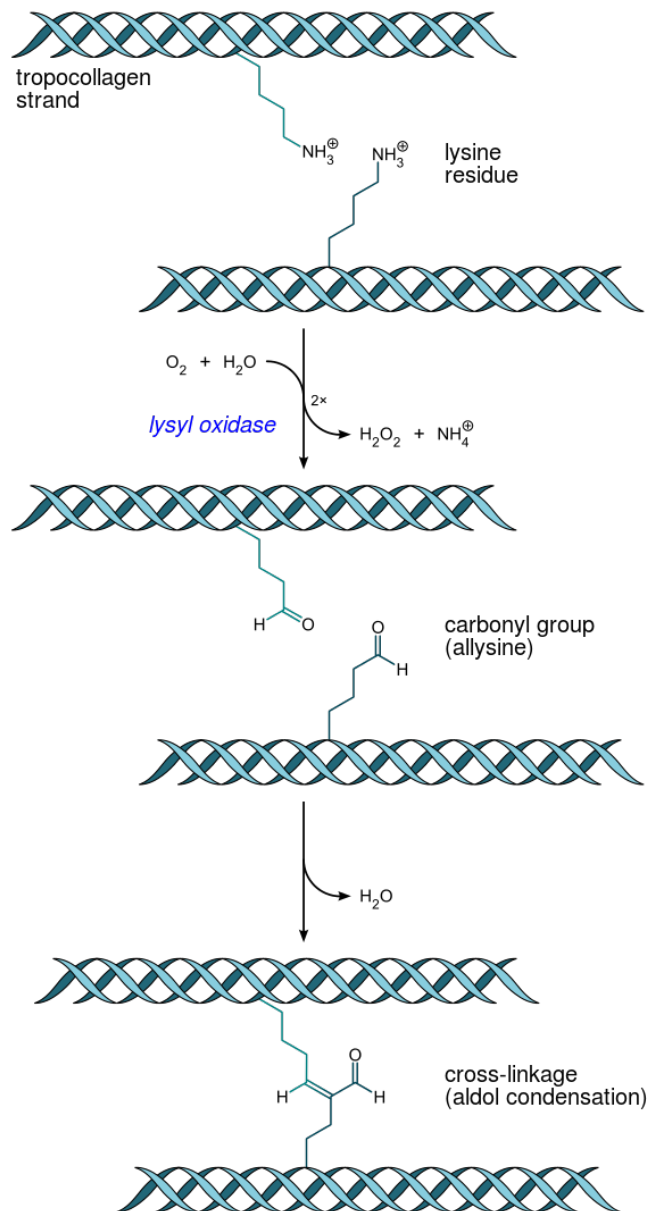
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Collagen – chemical crosslinking

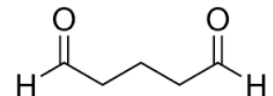


Dialdehydes have been long known in the leather industry as effective tanning agents and in histological laboratories as useful fixatives. Both of these applications are based on the reaction between **the dialdehyde and the ϵ -amino group of lysyl residues in the protein**, which induces formation of interchain cross-links.

Glutaraldehyde cross-linking is a relatively widely used procedure in the preparation of implantable biomaterials.

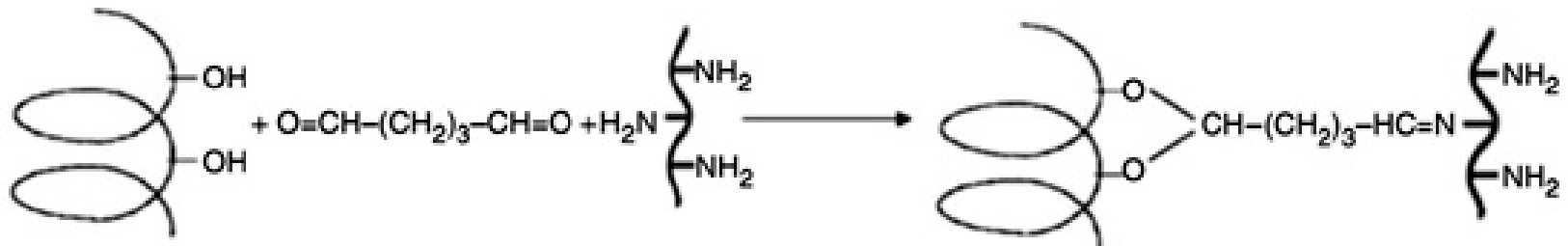
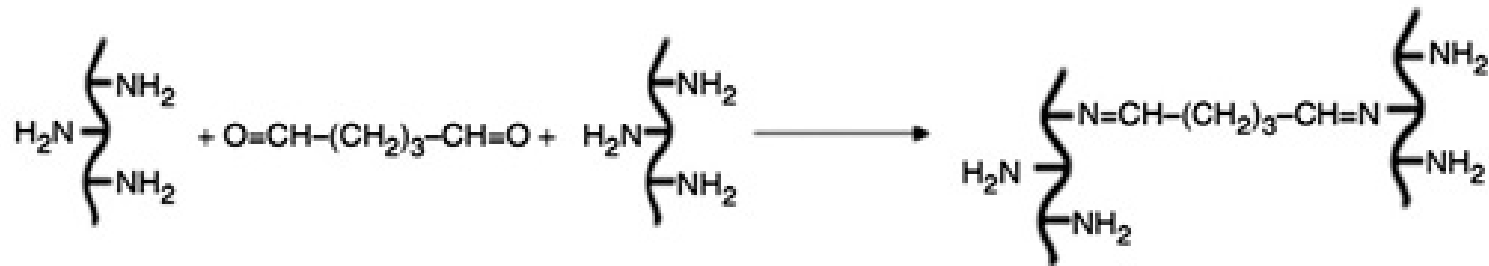
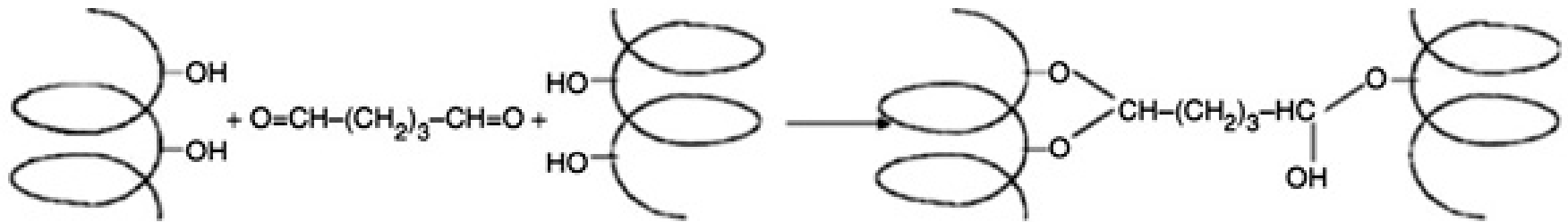
Free glutaraldehyde is a toxic substance for cells; it cross-links vital cell proteins. However, clinical studies and extensive clinical use of implants have shown that the toxicity of glutaraldehyde becomes effectively negligible after the unreacted glutaraldehyde has been carefully rinsed out following reaction with an implant, e.g., one based on collagen.

The nature of the cross-link formed has been the subject of controversy, primarily due to the complex, apparently polymeric, character of this reagent.

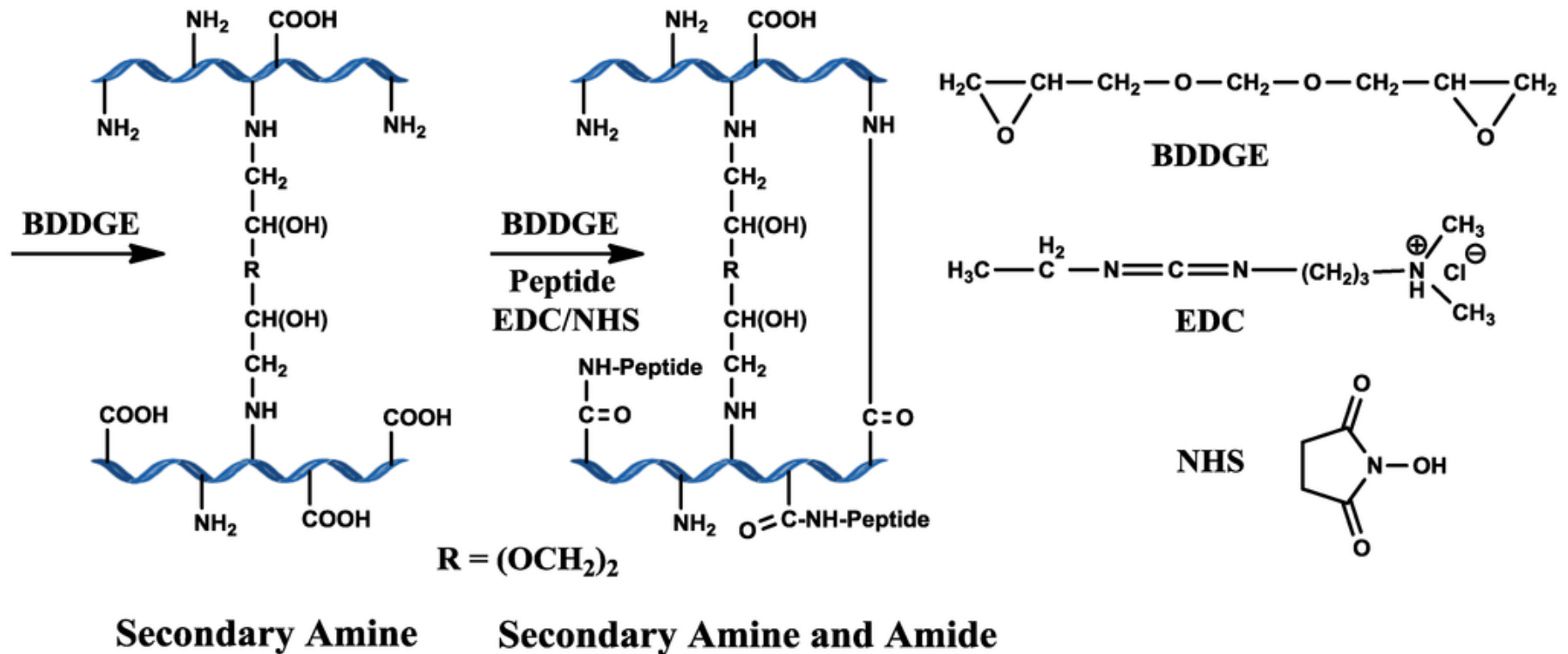


Collagen – chemical crosslinking

Glutaraldehyde for crosslinking



Collagen – chemical modification



Chemical cross-linking improved the elasticity and tensile strength of the collagen implants.

Collagen-based biomaterials – applications

Application	Physical state
Sutures	Extruded tape (Schmitt, 1985)
Hemostatic agents	Powder, sponge, fleece (Stengel <i>et al.</i> , 1974; Chvapil, 1979)
Blood vessels	Extruded collagen tube, processed human or animal blood vessel (Nimni, 1988)
Heart valves	Processed porcine heart valve (Nimni, 1988)
Tendon, ligaments	Processed tendon (Piez, 1985)
Burn treatment (dermal regeneration)	Porous collagen-glycosaminoglycan (GAG) polymer ^a (Yannas <i>et al.</i> , 1981, 1982, and 1989)
Peripheral nerve regeneration	Porous collagen-GAG copolymer (Chang and Yannas, 1992)
Meniscus regeneration	Porous collagen-GAG copolymers (Stone <i>et al.</i> , 1989)
Intradermal augmentation	Injectable suspension of collagen particles (Piez, 1985)
Gynecological applications	Sponges (Chvapil, 1979)
Drug-delivery systems	Various forms (Stenzel <i>et al.</i> , 1974, Chvapil, 1979)

^aSee also Chapter 7.10.